

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)		$\frac{38\,000\,000}{3\,750\,000}$ (1) Payback = 10.1 [years] (1) Accept 10 [years]		2		2	2	
		(ii)		Cost of {electricity / biomass / wood / energy} may change OR Energy use may change OWTTE Accept bills could change or savings could change		1		1		
	(b)			Growing trees (1) {removes / absorbs} CO ₂ from the atmosphere (1) [so disagree] Alternative: A tree absorbs as much CO ₂ (1) as it releases (1)			2	2		
	(c)	(i)		3 + 13.4 = 16.4 [MW] or 20.0 – 3.6 = 16.4 [MW]		1		1	1	
		(ii)		Efficiency = $\frac{16.4 \text{ ecf}}{20.0} \times 100$ (1) Efficiency = 82% (1) so disagree Conclusion must be present to award both marks Alternative: $\frac{15}{100} \times 20$ (1) Useful output = 3 M[W] (1) so disagree Conclusion must be present to award both marks			2	2	2	